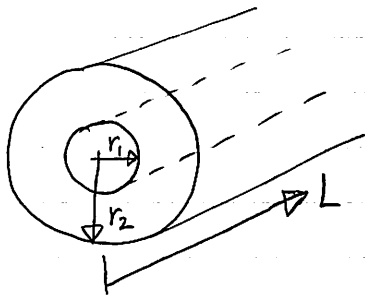


③ Concentric Cylinders

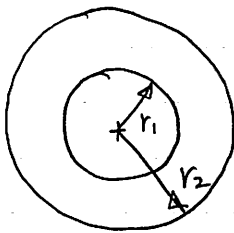


$$F_{12} = 1$$

$$\frac{A_1}{A_2} = \frac{r_1}{r_2}$$

$$q_{1-2} = \frac{2\pi r_1 L (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{r_1}{r_2} \left(\frac{1}{\epsilon_2} - 1 \right)}$$

④ Concentric Spheres

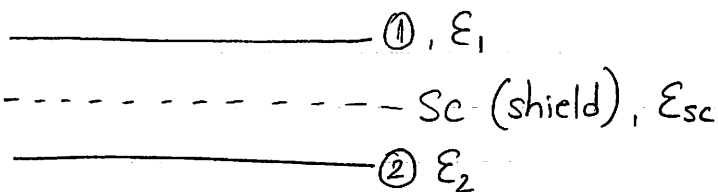


$$\frac{A_1}{A_2} = \frac{r_1^2}{r_2^2} ; F_{12} = 1$$

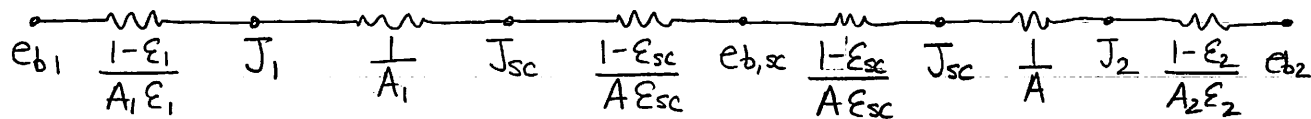
$$q_{1-2} = \frac{4\pi r_1^2 (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{r_1^2}{r_2^2} \left(\frac{1}{\epsilon_2} - 1 \right)}$$

Radiation Shields

They are often used to reduce heat losses



Drawing out our resistance diagram



$$q_{12} = \frac{A(e_{b1} - e_{b2})}{\left(\frac{1}{\epsilon_1} - 1 \right) + 1 + \left(\frac{1}{\epsilon_2} - 1 \right) + 2 \left(\frac{1}{\epsilon_{sc}} - 1 \right) + 1}$$

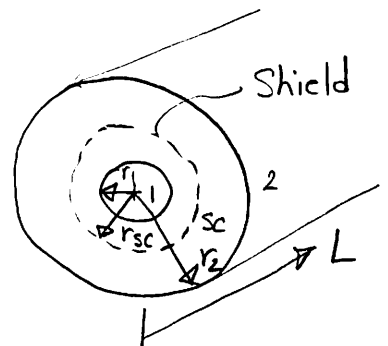
$$q_{12} = \frac{A(e_{b1} - e_{b2})}{\left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1\right) + \left(\frac{2}{\epsilon_{sc}} - 1\right)} \Rightarrow \text{For 1 shield}$$

For n shields in series:

$$q_{12} = \frac{A(e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} + \frac{2n}{\epsilon_{sc}} - (n+1)}$$

For cylindrical shields (n of them)

$$q_{12} = \frac{2\pi r_1 L (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{r_1}{r_2} \left(\frac{1}{\epsilon_2} - 1\right) + 2 \left(\frac{1}{\epsilon_{sc}} - 1\right) \sum_{j=1}^n \frac{r_1}{r_{sc,j}}}$$

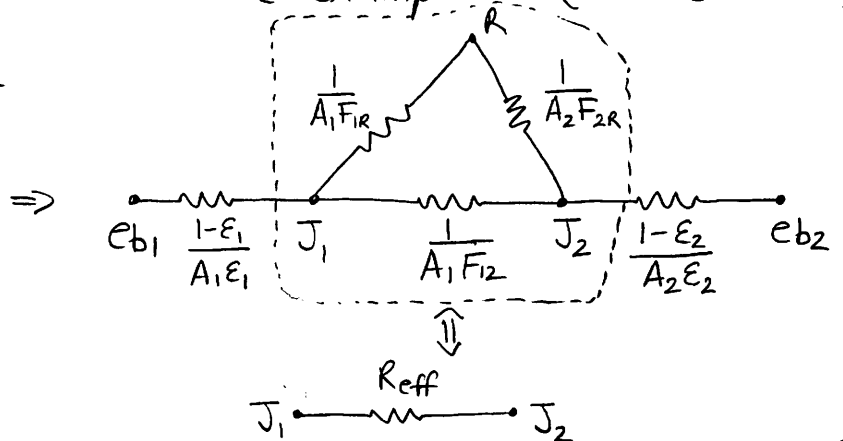
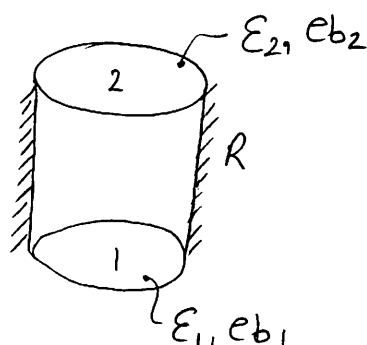


For spherical shields (n of them)

$$q_{12} = \frac{4\pi r_1^2 (e_{b1} - e_{b2})}{\frac{1}{\epsilon_1} + \frac{r_1^2}{r_2^2} \left(\frac{1}{\epsilon_2} - 1\right) + 2 \left(\frac{1}{\epsilon_{sc}} - 1\right) \sum_{j=1}^n \frac{r_1^2}{r_{sc,j}^2}}$$

Note, for spherical & cylindrical shields, you want r_{sc} to be close to r_1 since you want the shield to see as little of itself as possible.

Looking back at our adiabatic example: ($e_b = J$ at R)

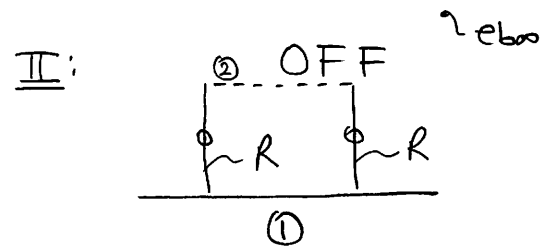
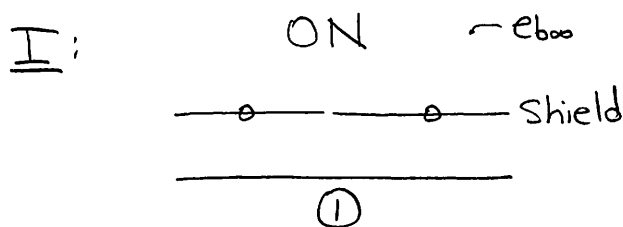


$$\frac{1}{R_{eff}} = \frac{1}{\frac{1}{A_1 F_{1R}} + \frac{1}{A_2 F_{2R}}} + A_1 F_{12}$$

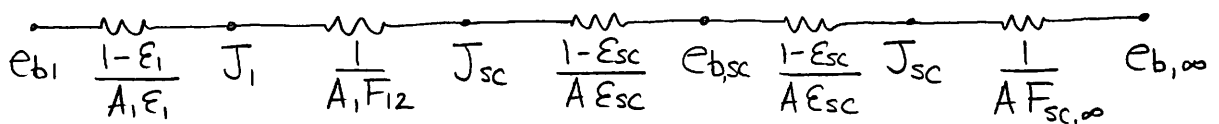
$$q_{12} = \frac{e_{b1} - e_{b2}}{\frac{1}{A_1} \left(\frac{1}{\epsilon_1} - 1 \right) + \frac{1}{A_2} \left(\frac{1}{\epsilon_2} - 1 \right) + R_{eff}}$$

⇒ Note, our solution does not depend on ϵ_R since $e_{bR} = J_R$ or $q'' = 0$ at R .

Example Consider a rotatable radiation shield shown below. Calculate its effectiveness in the on & off positions:

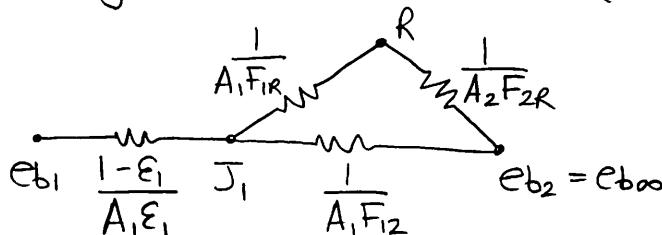


Solving the on case first:



$$q_{1\infty} = \frac{A_1 (e_{b1} - e_{b\infty})}{\frac{1}{\epsilon_1} + \frac{2}{\epsilon_{sc}} - 1}, \quad F_{sc, \infty} = 1$$

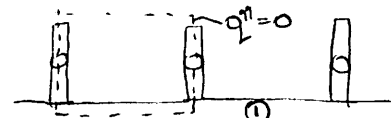
Looking at the OFF case (surfaces are reflective since $q'' = 0$ at the shields)



$$q_{12} = \frac{A_1 (e_{b1} - e_{b\infty})}{\frac{1}{\epsilon_1} - 1 + \sqrt{2}}$$

$$\left. \begin{aligned} F_{12} &= \sqrt{2} - 1 \\ F_{1R} &= 2 - \sqrt{2} \end{aligned} \right\} \text{From tables}$$

Aside: Note, the shields in II are adiabatic due to symmetry



$R_I \gg R_{II}$, Assuming $\epsilon_1 = 0.5$; $R = \text{thermal radiative resistance.}$
 $\epsilon_{sc} = 0.1$

$$\frac{R_I}{R_{II}} = \frac{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_{sc}} - 1}{\frac{1}{\epsilon_1} - 1 + \sqrt{2}} = 13.2$$

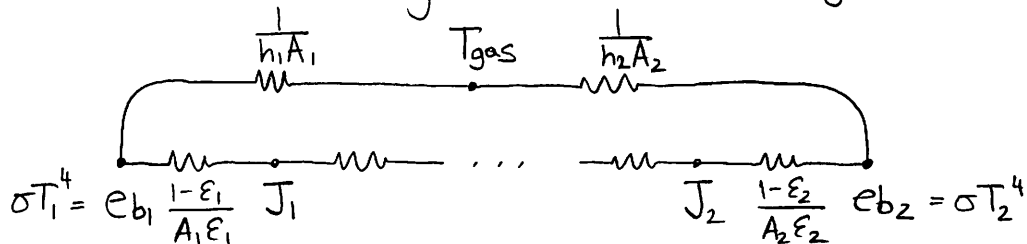
So when the shield is in the ON position, it is 13.2 times better at limiting heat transfer!

Multi-Mode Heat Transfer (Radiation + Convection)

Assuming we have radiation and convection for example, our analysis is similar:

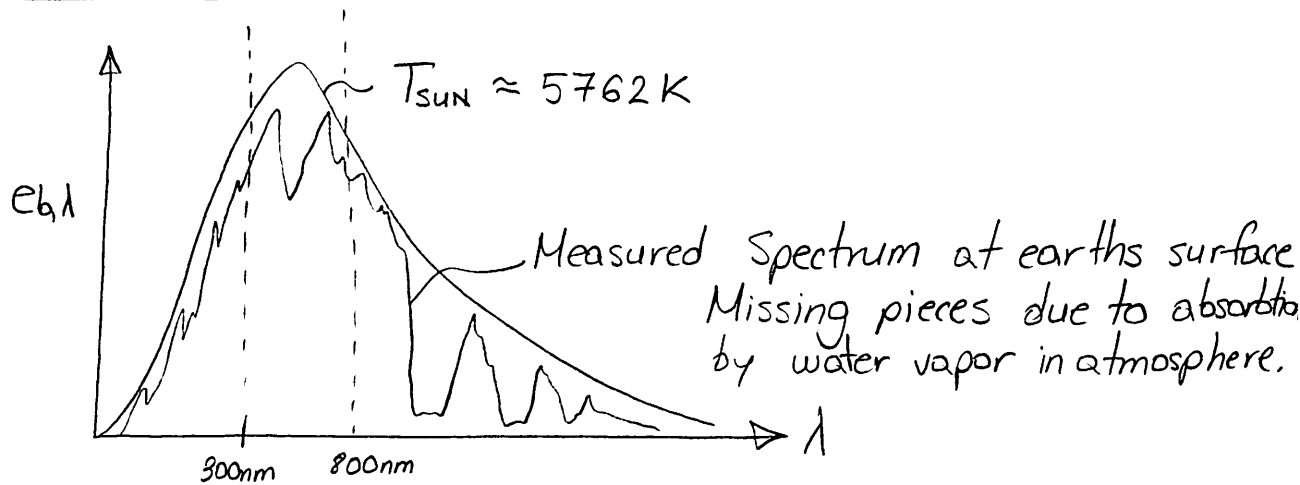


Our resistance diagram becomes: (T_{gas} is unknown)



We would have to solve the complete diagram, and it may require iteration, if T_1 or T_2 are not provided and instead q'' is provided.

Solar Radiation



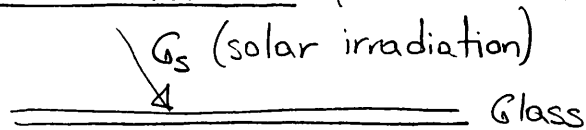
$$q_{\text{sun}} = 1353 \text{ W/m}^2 \text{ (Outside the atmosphere)}$$

$$q_{\text{earth}} = 636 \text{ W/m}^2 \text{ (On the earth's surface)}$$

Note, the total arriving energy from the sun to the earth is $\approx 1.7 \times 10^{14} \text{ kW}$. The US peak demand is $\approx 1 \times 10^9 \text{ kW}$.

Our eyes are most sensitive to visible light (400-700nm) where the peak intensity from the sun is. This is simply due to evolution.

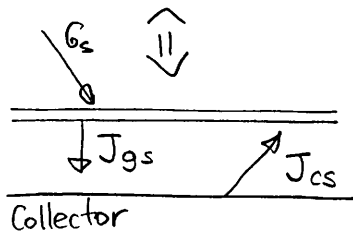
Solar Collectors: (Detailed Analysis)



glass: $\tau_{gs}, \rho_{gs}, \alpha_{gs}$

collector: ρ_{cs}, α_{cs}

Collector



Reflected energy J_{cs} from glass back to collector.

$$\begin{aligned} J_{gs} &= G_s \tau_{gs} + \rho_{gs} J_{cs} \\ J_{cs} &= \rho_{cs} J_{gs} \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{Reflected energy from glass}$$

$$J_{gs} = G_s \tau_{gs} + \rho_{gs} \rho_{cs} J_{gs}$$

$$J_{gs} = \frac{G_s \tau_{gs}}{1 - \rho_{gs} \rho_{cs}}$$

$$Q_{\text{collector},s} = \alpha_{cs} J_{gs} = G_s \frac{\tau_{gs} \alpha_{cs}}{1 - \rho_{gs} \rho_{cs}} \Rightarrow \text{Heat Collected in the solar spectrum.}$$

So now if we examine the IR range:

$$Q_{\text{coll-glass}} = \frac{e_{bc} - e_{bg}}{\frac{1}{\epsilon_c} + \frac{1}{\epsilon_g} - 1} \quad (\text{Two parallel plates})$$

If we look at the glass heat input: (From the solar spectrum)

$$\underbrace{\alpha_{gs}}_{\text{solar absorptivity}} (G_s + J_{cs}) = \alpha_{gs} G_s \left[1 + \frac{\rho_{cs} \tau_{gs}}{1 - \rho_{gs} \rho_{cs}} \right]$$

We also have heat input from the collector to the glass:

$$\alpha_{gs} G_s \left[1 + \frac{\rho_{cs} \tau_{gs}}{1 - \rho_{gs} \rho_{cs}} \right] + \frac{e_{bc} - e_{bg}}{\frac{1}{\epsilon_c} + \frac{1}{\epsilon_g} - 1} = \epsilon_g (e_{bg} - e_{b\infty}) + h(T_g - T_\infty)$$

↳ Energy balance at the glass.

Solving the above equations can give us our collector efficiency. Note here we had to split up the two spectral ranges since $\epsilon = \text{constant} = f(\lambda)$

Current collectors use micro/nanotechnology to achieve high absorptivity in the solar spectrum ($300\text{nm} < \lambda < 2500\text{nm}$) and low thermal emissivity in the IR spectrum ($\lambda > 2500\text{nm}$).